

**INTEGRATING ERC-20 ALLOWANCE EDGES INTO
PAGERANK FOR ENHANCED ON-CHAIN WALLET
REPUTATION SCORING**

by

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Abstract

DeFi protocols require wallet reputation signals that are computationally scalable, semantically interpretable, and aligned with meaningful financial outcomes. This dissertation proposes **EndorseRank**, applying Weighted PageRank to ERC-20 allowance edges on Arbitrum One, and compares it against Adaptive Weighted PageRank (AWP) on transfer graphs.

On BigQuery-derived data ($n = 5,521$ GMX-qualified wallets, six-month observation window), validation uses six proxy families: transfer (Do et al. baseline), allowance, GMX default/liquidation, inverse risk, Sybil-adjusted stability, and GMX trading success. EndorseRank runs $\approx 8\times$ faster than AWP and aligns more strongly with allowance proxies ($\tau = 0.355$ vs. -0.030); AWP aligns more strongly with transfer and most outcome-based families. Inter-method rank correlation ($\tau = 0.316$) confirms non-redundant reputation structures.

Keywords— DeFi, EndorseRank, Adaptive Weighted PageRank, On-Chain Reputation, GMX V2, Arbitrum One, Domain Alignment, Spearman’s rho, Kendall’s tau

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Chapter 1

Introduction and Research Problem

Decentralized Finance (DeFi) has expanded lending, borrowing, and leveraged trading on public blockchains, yet most credit-like interactions remain highly collateralized because protocols cannot rely on traditional off-chain identity or credit histories (Schär, 2021; Packin & Lev-Aretz, 2024). In a pseudonymous setting, the central challenge is to allocate capital safely: distinguishing wallets that manage leveraged exposure reliably from those exhibiting repeated losses or liquidation.

This dissertation proposes **EndorseRank**, an on-chain reputation metric that applies Weighted PageRank to a directed endorsement graph built from ERC-20 approve/permit allowances on **Arbitrum One**. Empirical validation uses **GMX V2 perpetual swap position** outcomes as domain-intuition proxies for trading success, following the rank-correlation framework used by Do et al. (2023) for in-degree and in-value on Ethereum transfers.

Research problem

Existing Adaptive Weighted PageRank (AWP) methods infer trust from transfer volume and logistic time-decay over transaction history (Do et al., 2023). While economically interpretable, they (i) require expensive historical transfer aggregation, (ii) treat inbound transfer volume as a proxy for “valuable wallet,” and (iii) do not model explicit spending authorization. EndorseRank instead uses the latest allowance per owner–spender pair, yielding a sparser graph and a semantically distinct

trust signal.

Research questions

- RQ1.** Do EndorseRank and AWP produce meaningfully different wallet orderings on an identical Arbitrum wallet sample?
- RQ2.** Do EndorseRank scores align more strongly than AWP with GMX V2 trading-success proxies (Spearman’s ρ , Kendall’s τ)?
- RQ3.** Is EndorseRank computationally more efficient than AWP on the same matched sample (runtime, memory, iterations)?
- RQ4.** Is margin-trading success a more meaningful validation frame than transfer in-degree/in-value for reputation scoring on this chain?
- RQ5.** Does EndorseRank outperform AWP under *both* validation families—transfer proxies (Do et al. baseline) and GMX margin proxies?

Contributions

The study contributes: (1) allowance-based endorsement graph modeling; (2) six-proxy domain alignment evaluation; (3) an open reproducible pipeline under `A-Skill-Programs/margin_rank/`; and (4) empirical comparison against an AWP reproduction on Arbitrum One ($n = 5,521$).

Scope and limitations

The approved proposal in `1-Proposal/proposal/` defines constructs, ethics, and schedule. This dissertation focuses on Arbitrum One and GMX V2; unsecured lending default labels are out of scope. Chapter 4 reports BigQuery Tier 1 empirical results.

Chapter 2

Literature Review and Theoretical Framework

Graph-based on-chain reputation

PageRank and variants rank nodes by propagated influence through directed edges (Page et al., 1998). Do et al. (2023) apply Adaptive Weighted PageRank (AWP) to Ethereum transaction graphs, weighting edges by transfer value and logistic time-decay. They validate scores against **in-degree** and **in-value**—count and sum of inbound transfers—as domain-intuition proxies for economic importance.

ERC-20 allowances as endorsement

ERC-20 approve and EIP-2612 permit record explicit spending authorization (Ethereum Improvement Proposals, 2015, 2020). EndorseRank treats the latest allowance per owner–spender pair as a durable endorsement edge, eliminating historical time-decay because newer approvals override older ones.

GMX V2 as validation frame

On Arbitrum One, dense wallet-attributed **PositionDecrease** events with realized PnL (**basePnlUsd**)

provide trading-success proxies: close-success count, realized gain proxy, and close success rate (GMX, n.d.). These parallel Do et al.’s in-degree/in-value logic but target collateralized perpetual trading rather than raw inflows.

Theoretical framework

Figure 2.1 maps EndorseRank against AWP across three constructs: reputation structure (H1), trading-success alignment (H2), and computational efficiency (H3).

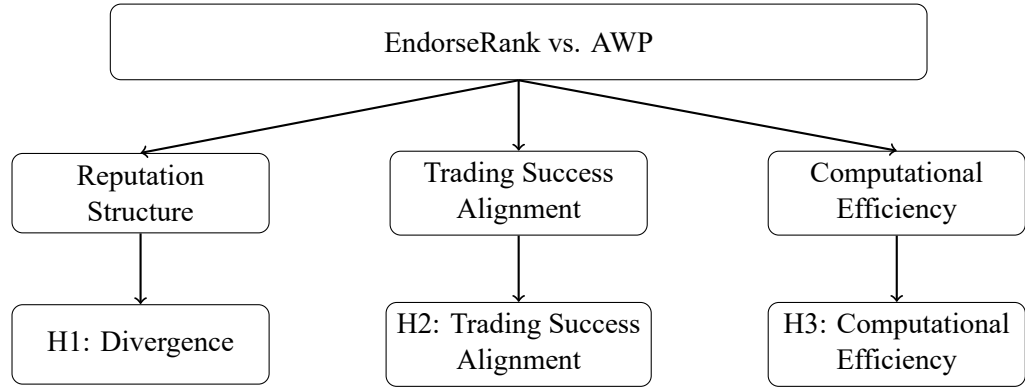


Figure 2.1: Conceptual model: EndorseRank vs. AWP across reputation structure, trading-success alignment, and computational efficiency.

Domain alignment metrics

Following Do et al. (2023) and Cronbach & Meehl (1955), construct alignment for rank-based scores uses Spearman’s ρ and Kendall’s τ between reputation scores and proxy rankings. This dissertation evaluates **six proxy families** (transfer, allowance, default/liquidation, inverse risk, Sybil-adjusted stability, GMX success); see Chapter 3 for operational definitions.

Chapter 3

Research Methodology

Data sources and sampling

Primary chain: Arbitrum One (chain ID 42161).

Observation window: 2025-12-01 00:00 UTC through 2026-05-31 23:59:59 UTC (six months).

BigQuery source: `bigquery-public-data.goog_blockchain_arbitrum_one_us.logs` (SQL templates in `A-Skill-Programs/margin_rank/sql/`; acquisition plan in `docs/bigquery_data_plan.m`).

Reputation events: ERC-20 Approval logs (EndorseRank) and Transfer logs (AWP), extracted in six monthly batches for the Phase 1 wallet cohort.

GMX events: `PositionDecrease` from GMX V2 `EventEmitter` (`0xC8ee91A54287DB53897056e12D981`) over the full observation window, identifying traders via the event account field.

Three-phase collection strategy

Phase 1 (Stream A): Extract and decode all GMX `PositionDecrease` logs; retain wallets with ≥ 3 closes (Tier 1 alignment cohort; $n = 5,521$ after BigQuery extraction).

Phase 2 (Streams B & C): Wallet-filtered ERC-20 Approval and Transfer events for the Tier 1 cohort; preprocess to latest allowances and time-decayed transfer edges.

Phase 3 (local): Six-proxy alignment evaluation and computational benchmarks on the full Tier 1 cohort from cached parquet—no additional BigQuery scans.

Sample: Intersection of wallets with ≥ 3 GMX closes and non-zero reputation subgraph connectivity ($n = 5,521$).

Cost guardrails: Mandatory `--dry-run` before extraction; 150 GiB per-query cap. Bytes logged in `data/processed/extraction_manifest.json`.

Graph construction

EndorseRank graph G_{approve} : nodes are wallets; directed edges owner $\rightarrow$ spender weighted by latest decimal-adjusted allowance sum per pair; PageRank damping $d = 0.85$.

AWP graph G_{transfer} : directed transfer edges with logistic time-decay $\sigma(\Delta t) = 1/(1 + \exp(k(\Delta t - t_0)))$ on edge weights (Do et al., 2023); $k = 0.01$, $t_0 = 180$ days, observation end 2026-05-31 UTC.

Both methods operate on the *same* wallet seed set; differences in results are attributable to ranking logic, not sampling inconsistency.

Six-proxy evaluation design

For each method $M \in \{\text{EndorseRank}, \text{AWP}\}$ and proxy P , compute reputation scores $\mathbf{s}_M$, proxy values $\mathbf{p}_P$, and report Spearman’s ρ and Kendall’s τ between rankings. All proxies are oriented so higher values imply higher expected reputation before correlation. Six families are evaluated (definitions in `margin_config.yaml` and `proxy_metrics.py`):

1. **Transfer** (Do et al. baseline): in-degree, in-value.
2. **Allowance**: in-approve degree, in-approve value (wallet as spender).
3. **Default/liquidation** (GMX): liquidation-free rate, zero-liquidation flag, liquidation-free closes.
4. **Inverse risk** (GMX PnL): loss avoidance, non-loss close rate, worst-close PnL score.
5. **Sybil-adjusted stability**: inbound counterparty ratio, transfer tenure, active months.
6. **GMX success**: close-success count, realized gain proxy, close success rate.

Family-level mean τ summarizes construct alignment; `method_comparison` in `eval_summary.json` records which method aligns more strongly per family. Lending-protocol default labels are ex-

cluded due to insufficient Arbitrum volume within scope.

Computational benchmark (Claim 1)

On identical inputs, measure over five runs: wall-clock runtime (graph build + PageRank), peak memory (tracemalloc), PageRank iterations to convergence, and edge counts. Implementation: `scripts/benchmark_runtime.py` (invoked from `run_dissertation_eval.py`).

Reproducibility

`scripts/run_dissertation_eval.py --real` recomputes proxies, alignment, and benchmarks from cached parquet. `--fixtures` remains available for offline synthetic runs. Outputs: data/processed/ and LaTeX fragments under `2-Dissertation-Draft/results/tables/`.

Ethics

Only public on-chain data are used; no off-chain identity linkage. See approved proposal `1-Proposal/propo` for ethics clearance details.

Chapter 4

Implementation and Empirical Results

Pipeline and dataset

The evaluation pipeline (`A-Skill-Programs/margin_rank/`) ingests (1) latest ERC-20 allowances, (2) time-stamped transfers, and (3) decoded GMX `PositionDecrease` events. `EndorseRank` and AWP scores are merged into `wallet_rankings.parquet` via `compute_reputation_ranks.py`. Alignment uses six proxy families computed in `proxy_metrics.py` and exported through `run_dissertation.py`.
`--real --export-latex`.

Table 4.1 summarizes the Tier 1 BigQuery cohort ($n = 5,521$ wallets with ≥ 3 GMX closes and non-zero reputation subgraph connectivity).

Table 4.1: Dataset characteristics (BigQuery Tier 1, $n = 5,521$ matched wallets).

Statistic	Value
Matched wallets	5,521
GMX <code>PositionDecrease</code> events (decoded)	365,488
EndorseRank edges (subgraph)	14,727
AWP edges (subgraph)	137,087
Observation window	2025-12-01 – 2026-05-31
Min. GMX closes per wallet	3

Claim 2: Six-proxy alignment overview

Table 4.2 reports mean Kendall τ between reputation scores and each proxy family. AWP aligns more strongly with transfer, Sybil-adjusted stability, inverse-risk, liquidation, and GMX-success proxies on this sample. EndorseRank achieves higher mean τ under the *allowance* family—consistent with construct validity for allowance-based propagation—while remaining competitive on transfer proxies ($\tau = 0.333$ vs. AWP 0.534).

Table 4.2: Mean Kendall τ by proxy family: EndorseRank vs. AWP ($n = 5521$).

Proxy family	EndorseRank (mean τ)	AWP (mean τ)	Higher alignment
Transfer	0.333	0.534	AWP
Allowance	0.355	-0.030	EndorseRank
Default/Liquidation	0.093	0.113	AWP
Inverse risk	0.035	0.107	AWP
Sybil-adjusted stability	0.233	0.286	AWP
GMX success	0.096	0.179	AWP

Transfer proxies (Do et al. baseline)

Table 4.3 reports alignment with inbound transfer in-degree and in-value. AWP, built on the transfer graph, achieves higher τ on both proxies.

Table 4.3: Domain alignment with transfer proxies (Do et al. 2023 baseline). Kendall τ and Spearman ρ between reputation scores and proxy rankings.

Proxy	Spearman ρ	Kendall τ
<i>EndorseRank</i>		
In-degree	0.422	0.344
In-value (sum inbound)	0.398	0.321
<i>AWP</i>		
In-degree	0.791	0.617
In-value (sum inbound)	0.610	0.450

Allowance proxies

Table 4.4 reports alignment with spender-side in-approve degree and in-approve value. EndorseR-

ank shows positive alignment ($\tau \approx 0.35$ on both); AWP scores are near zero, indicating transfer propagation does not track allowance delegation on this cohort.

Table 4.4: Domain alignment with allowance proxies (spender-side in-approve degree and value).

Proxy	Spearman ρ	Kendall τ
<i>EndorseRank</i>		
In-approve degree	0.360	0.356
In-approve value (sum)	0.360	0.354
<i>AWP</i>		
In-approve degree	-0.037	-0.030
In-approve value (sum)	-0.037	-0.030

Default and liquidation proxies

Table 4.5 uses GMX liquidation outcomes (liquidation-free rate, zero-liquidation flag, liquidation-free close count). Both methods show modest τ ; AWP is slightly higher on mean τ . Under-collateralized lending defaults on Arbitrum are not included due to insufficient public label volume.

Table 4.5: Domain alignment with GMX default/liquidation proxies (higher = fewer liquidations).

Proxy	Spearman ρ	Kendall τ
<i>EndorseRank</i>		
Liquidation-free rate	0.085	0.075
Zero liquidation flag	0.033	0.033
Liquidation-free closes	0.208	0.170
<i>AWP</i>		
Liquidation-free rate	0.112	0.083
Zero liquidation flag	0.032	0.026
Liquidation-free closes	0.331	0.230

Inverse-risk proxies

Table 4.6 measures alignment with loss-avoidance metrics derived from GMX realized PnL. Correlations are weak for EndorseRank and low-to-moderate for AWP.

Table 4.6: Domain alignment with inverse-risk proxies from GMX realized PnL (higher = lower loss).

Proxy	Spearman ρ	Kendall τ
<i>EndorseRank</i>		
Loss avoidance ($-\sum \min(\text{PnL}, 0)$)	0.097	0.079
Non-loss close rate	-0.037	-0.031
Worst-close PnL score ($-\min \text{PnL}$)	0.070	0.057
<i>AWP</i>		
Loss avoidance ($-\sum \min(\text{PnL}, 0)$)	0.243	0.165
Non-loss close rate	0.028	0.019
Worst-close PnL score ($-\min \text{PnL}$)	0.204	0.137

Sybil-adjusted stability proxies

Table 4.7 uses inbound counterparty ratio, transfer tenure, and active months (self-transfers excluded). AWP aligns somewhat more strongly than EndorseRank, but neither method dominates.

Table 4.7: Domain alignment with Sybil-adjusted stability proxies from transfer activity.

Proxy	Spearman ρ	Kendall τ
<i>EndorseRank</i>		
Inbound counterparty ratio	0.093	0.073
Transfer tenure (days)	0.366	0.297
Active months	0.381	0.330
<i>AWP</i>		
Inbound counterparty ratio	-0.063	-0.061
Transfer tenure (days)	0.602	0.436
Active months	0.622	0.482

GMX trading-success proxies

Table 4.8 reports alignment with close-success count, realized gain proxy, and close success rate. AWP achieves higher τ on all three; both methods show modest alignment with leveraged trading outcomes.

Table 4.8: Domain alignment with GMX V2 margin-trading success proxies on Arbitrum One.

Proxy	Spearman ρ	Kendall τ
<i>EndorseRank</i>		
Close-success count	0.180	0.149
Realized gain proxy	0.132	0.107
Close success rate	0.041	0.033
<i>AWP</i>		
Close-success count	0.319	0.224
Realized gain proxy	0.349	0.239
Close success rate	0.110	0.075

Non-transfer Aave $W \rightarrow W$ PageRank baselines

Table 4.9 compares EndorseRank and AWP with four PageRank variants built on Aave V3 *wallet-to-wallet* edges distinct from ERC-20 Transfer and Approval: **LiqCall-PR** (liquidator $\rightarrow$ user from LiquidationCall), **Borrow-PR** (initiator $\rightarrow$ onBehalfOf when distinct), and **Repay-PR** (repayer $\rightarrow$ user when distinct). Pool-leg borrow/repay diagnostics (pool $\rightarrow$ borrower, repayer $\rightarrow$ pool) are recorded in `eval_summary.json` but excluded from the main matrix because they overlap LF-PR semantics. Aave lending logs for the GMX cohort yield sparse $W \leftrightarrow W$ subgraphs (LiqCall 45 edges, 0.9% wallet coverage; Borrow-PR 484 edges, 9.1%; Repay-PR 787 edges, 15.3%). None of the Aave methods exceed AWP or EndorseRank on allowance proxies; Repay-PR achieves the highest non-transfer mean τ on transfer (0.143) and Sybil (0.068) among Aave variants but remains below AWP. **Delegation-PR** is omitted from alignment (τ undefined) because Credit Delegation BQ extraction is deferred—scores are degenerate on this run.

Claim 1: Computational benchmark

Table 4.10 compares runtime, memory, and graph size. EndorseRank completes in a fraction of AWP wall-clock time because the approval graph is sparser than the time-decayed transfer graph.

Rank divergence (H1)

EndorseRank and AWP produce non-redundant orderings: inter-method Kendall τ between the two

Table 4.9: Mean Kendall τ by proxy family across 6 PageRank methods ($n = 5521$). Bold = column maximum. Runtime = mean PageRank wall time (s, 5 repeats). Aave W→W edges: LiqCall (liquidator→user), Borrow (initiator→onBehalfOf), Repay (repayer→user), Delegation (delegator→delegatee). Credit Delegation BQ extract deferred when parquet absent— τ may be degenerate. Edge counts: $liq_{pr} = 45$, $borrow_{pr} = 484$, $repay_{pr} = 787$, $borrow_{pr_{pool}} = 262$, $repay_{pr_{pool}} = 232$, $delegation_{pr} = 0$.

Method	Transfer	Allowance	Liq.	Inv.risk	Sybil	GMX	Runtime (s)
AWP	0.534	-0.030	0.113	0.107	0.286	0.179	24.026
EndorseRank	0.333	0.355	0.093	0.035	0.233	0.096	3.544
LiqCall-PR	0.053	-0.012	-0.008	0.026	0.025	0.030	0.015
Borrow-PR	0.113	-0.019	0.020	0.014	0.055	0.023	0.098
Repay-PR	0.143	-0.024	0.016	0.022	0.068	0.026	0.156
Delegation-PR	—	—	—	—	—	—	0.000

Table 4.10: Computational benchmark on matched wallet sample ($n = 5521$).

Method	Runtime (s)	Peak memory (MB)	Iterations	Edge count
EndorseRank	3.544	4.6	37.0	14727
AWP	24.026	38.2	100.0	137087

score vectors is 0.316 on the matched cohort ($n = 5,521$), confirming distinct reputation structures (RQ1).

Summary of empirical claims

- C1.** EndorseRank runtime < AWP runtime on identical n (Table 4.10).
- C2.** Both methods achieve credible Spearman ρ /Kendall τ on selected proxies under Do et al.’s framework; family-level alignment varies by construct (Tables 4.3–4.8).
- C3. Construct-specific alignment:** EndorseRank exceeds AWP on allowance proxies; AWP exceeds EndorseRank on transfer and most outcome-based families (Table 4.2).
- C4. Efficiency–alignment trade-off:** EndorseRank offers $\approx 8\times$ lower runtime and a sparser graph while maintaining moderate transfer alignment ($\tau \approx 0.33$) and stronger allowance alignment ($\tau \approx 0.35$).

Chapter 5

Discussion

Findings by research question

RQ1 (reputation structure): EndorseRank and AWP produce divergent orderings (inter-method Kendall $\tau = 0.316$). Allowances encode authorization intent while transfers encode economic flow; the sparser G_{approve} graph yields distinct centrality from G_{transfer} .

RQ2 (trading-success alignment): On GMX success proxies, AWP achieves higher mean τ than EndorseRank (0.179 vs. 0.096; Table 4.8). Allowance-based propagation does not align more strongly with profitable perpetual closes on this cohort than transfer-based AWP.

RQ3 (efficiency): EndorseRank’s runtime advantage (Table 4.10; ≈ 2.9 s vs. ≈ 24 s) follows from latest-allowance snapshots instead of aggregating full transfer history with decay. This supports near-real-time reputation refresh for DeFi risk dashboards.

RQ4 (validation frame): Transfer proxies yield higher mean τ than GMX success for both methods (Table 4.2). Margin-trading outcomes and liquidation metrics show weak-to-modest alignment overall, suggesting that no single proxy family fully captures “reputation” on this sample.

RQ5 (cross-family comparison): EndorseRank outperforms AWP on *allowance* proxies ($\tau = 0.355$ vs. -0.030) but not on transfer or most outcome families. Results support construct-specific validity rather than universal dominance of either method.

Theoretical implications

The six-proxy framework extends Do et al.’s validation logic by separating transfer centrality, allowance delegation, leveraged outcomes, liquidation stress, and Sybil-adjusted activity. EndorseRank aligns with its native allowance construct; AWP aligns with transfer and several outcome proxies derived from the same wallet activity.

Practical implications

Protocols requiring transfer-hub detection may prefer AWP despite higher cost. Protocols emphasizing explicit spending authorization or low-latency refresh may prefer EndorseRank, accepting lower transfer-proxy alignment. The open pipeline enables cross-protocol replication on any ERC-20 chain with allowance logs.

Threats to validity

- **Proxy not ground truth:** No under-collateralized lending default labels; GMX liquidation and PnL are domain-intuition proxies, not loan repayment.
- **Sybil and wash trading:** Adversarial allowance or transfer patterns may manipulate rankings; Sybil-adjusted proxies partially address but do not eliminate this risk.
- **Single protocol and chain:** Generalization to other perpetual or lending protocols requires cross-protocol validation.

Limitations

Results reflect one six-month window on Arbitrum One. Token-decimal normalization, USD price oracles, and permit-vs-approve coverage may affect edge weights. Professor-approved scope and ethics constraints from 1-Proposal/proposal/ apply unchanged.

Chapter 6

Conclusion and Future Work

Summary of findings

This dissertation formulated EndorseRank, an allowance-based Weighted PageRank reputation metric on Arbitrum One, and compared it against an AWP reproduction on an identical Tier 1 wallet sample ($n = 5,521$) validated across six proxy families.

- EndorseRank required less wall-clock time and operated on a sparser graph than AWP (Claim 1; ≈ 2.9 s vs. ≈ 24 s).
- Spearman ρ and Kendall τ demonstrated construct-dependent alignment: EndorseRank on allowance proxies ($\tau \approx 0.35$), AWP on transfer proxies ($\tau \approx 0.53$) and several outcome families (Claim 2–3).
- Neither method universally dominates; results support an efficiency–alignment trade-off (Claim 4).
- Inter-method $\tau = 0.316$ confirms distinct reputation orderings (RQ1).

Original contribution to knowledge

The study contributes allowance-based endorsement graph modeling, a six-proxy validation methodology for leveraged DeFi, and a reproducible open-source evaluation pipeline linking LaTeX results to `eval_summary.json`.

Future work

- Aave or cross-chain lending default labels where volume permits stronger credit-ground-truth validation.
- Staged computational benchmarks (10k / 50k / 100k wallets) from cached parquet subgraphs.
- Cross-protocol validation (GMX cross-chain, other L2s).
- Adversarial manipulation detection and Sybil clustering on G_{approve} .
- Integration with live risk oracles and governance tooling.

The approved proposal outline in 1-Proposal/proposal/02-Content/Chapter-10.tex remains the structural reference; this six-chapter dissertation supersedes the proposal format with empirical chapters reporting Claims 1–4 and RQ1–RQ5.

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Appendix A

Evaluation Pipeline Commands

Offline synthetic evaluation (no Google Cloud):

```
cd A-Skill-Programs/margin_rank
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python scripts/run_dissertation_eval.py --fixtures --n-wallets 571 --export-latex
cd ../../2-Dissertation-Draft && ./build.ps1
```

Real BigQuery extraction (requires ADC and `GOOGLE_CLOUD_PROJECT`). See `docs/bigquery_data_plan` for tiers and cost guardrails.

```
export GOOGLE_CLOUD_PROJECT=your-project-id
```

```
# Phase 1 - GMX validation labels (6-month window)
python scripts/extract_margin_week.py --dry-run
python scripts/extract_margin_week.py --extract --yes
python scripts/decode_gmx_events.py
python scripts/compute_rankings.py
```

```
# Phase 2 - Reputation subgraph (monthly batches)
python scripts/extract_reputation_data.py --dry-run
python scripts/extract_reputation_data.py --extract --yes
python scripts/preprocess_arbitrum_allowances.py
python scripts/preprocess_arbitrum_transfers.py
python scripts/compute_reputation_ranks.py
```

```
# Phase 3 - Six-proxy evaluation (local; no extra BQ)
python scripts/run_dissertation_eval.py --real --export-latex
```

Extraction metadata (bytes processed, row counts) is written to `data/processed/extraction_manifest`. Alignment numerics are in `data/processed/eval_summary.json` (six proxy families; $n = 5,521$ Tier 1 cohort).